

Arthur W. Juliani, PhD

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Education

University of Oregon , Eugene, OR · <i>Ph.D. in Psychology</i>	2016 – 2020
– Dissertation: “Learning and Acting with Predictive Cognitive Maps”	
University of Oregon , Eugene, OR · <i>M.S. in Psychology</i>	2014 – 2015
North Carolina State University , Raleigh, NC · <i>B.A. in Psychology</i>	2010 – 2013

Research Experience

Institute for Advanced Consciousness Studies , <i>Senior Research Scientist</i>	2024 – Present
– Conducted research on social cognition and large language models, publishing papers on AI embodiment and consciousness.	
– Collaborated with an international research team to develop a theoretical framework for embodiment-based self- and other-care in simulated artificial agents.	
Microsoft Research , <i>Postdoctoral Research Fellow</i>	2022 – 2024
– Developed and maintained Neuro-Nav, an open-source educational library for teaching reinforcement learning concepts through neurally-plausible simulations.	
– Collaborated with research groups across Microsoft Research and external partners on machine learning theory, computational neuroscience, and psychedelic science.	
– Led a research project evaluating methods for addressing distributional shift and improving robustness in deep neural networks, resulting in a NeurIPS Spotlight paper.	
– Developed and maintained the Neuro-Nav open source library for deep reinforcement learning research in psychology and computational neuroscience.	
Araya Inc , <i>Senior Research Scientist</i>	2021 – 2022
– Mentored PhD interns through the full research lifecycle, developing training materials and providing individualized guidance on research design, methodology, and scientific communication.	
– Led a project to develop a novel model of the global neuronal workspace using a Transformer-like architecture, and validated the model on working memory and attentional control tasks.	
– Supervised intern projects on neural decoding of fMRI and ECoG data to identify neural correlates of the global workspace and brain-computer interface targets.	
– Developed a theoretical framework linking attentional information access in cognitive neuroscience to attention mechanisms in machine learning models.	
Unity Technologies , <i>Senior Research Engineer</i>	2017 – 2021
– Created ML-Agents Toolkit (18.9k GitHub stars), an open-source educational platform used by thousands of students and researchers to learn reinforcement learning through interactive 3D environments.	
– Founded and developed the ML-Agents Toolkit (18.9k GitHub stars), an open-source library for training deep reinforcement learning agents in virtual environments built with the Unity engine.	
– Implemented and maintained state-of-the-art deep reinforcement learning algorithms, including DRQN, PPO, GAIL, and ICM, in both TensorFlow and PyTorch.	
– Led development of Obstacle Tower, a procedurally generated 3D environment for evaluating RL generalization, and organized a public challenge attracting over 600 competing teams.	
University of Oregon , <i>Graduate Research Assistant</i>	2014 – 2016
– Led undergraduate research teams, providing hands-on mentorship in experimental design, data analysis, and scientific writing.	

- Proposed and evaluated novel neural network models of hippocampal function for spatial navigation and memory.
- Designed human behavioral and perceptual experiments deployed in 3D virtual environments, leading undergraduate research teams in conducting and publishing the studies.

Selected Publications

- Safron, A., **Juliani, A.**, Reggente, N., Klimaj, V., & Johnson, M. (2025). On the varieties of conscious experiences: altered beliefs under psychedelics (ALBUS). *Neuroscience of Consciousness*, 2025(1), niae038.
- Juliani, A.**, & Ash, J. (2024). A Study of Plasticity Loss in On-Policy Deep Reinforcement Learning. *Advances in Neural Information Processing Systems*, 37, 113884-113910. ***Spotlight Paper***
- Juliani, A.**, Safron, A., Kanai, R. (2024). Deep CANALs: A Deep Learning Approach to Refining the Canalization Theory of Psychopathology. *Neuroscience of Consciousness*.
- Juliani, A.**, Barnett, S., Davis, B., Sereno, M., & Momennejad, I. (2022). Neuro-Nav: a library for neurally-plausible reinforcement learning. In *Proceedings of the 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making*.
- Juliani, A.**, Arulkumaran, K., Sasai, S., & Kanai, R. (2022). On the link between conscious function and general intelligence in humans and machines. *Transactions On Machine Learning Research*. ***Survey Certification***

Additional Publications

- Dossa, R. F. J., Arulkumaran, K., **Juliani, A.**, Sasai, S., & Kanai, R. (2024). Design and evaluation of a global workspace agent embodied in a realistic multimodal environment. *Frontiers in Computational Neuroscience*, 18, 1352685.
- Milani, S., **Juliani, A.**, Momennejad, I., Georgescu, R., Rzepecki, J., Shaw, A., ... & Hofmann, K. (2023). Navigates Like Me: Understanding How People Evaluate Human-Like AI in Video Games. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*.
- Juliani, A.**, Kanai, R., & Sasai, S. S. (2022). The Perceiver Architecture is a Functional Global Workspace. In *Proceedings of the Annual Meeting of the Cognitive Science Society*.
- Juliani, A.**, (2020). *Learning and Acting with Predictive Cognitive Maps* [Doctoral dissertation, University of Oregon]. ProQuest Dissertation Publishing.
- Juliani, A.**, Khalifa, A., Berges, V. P., Harper, J., Teng, E., Henry, H., Crespi, A., Togelius, J., & Lange, D. (2019). Obstacle tower: A generalization challenge in vision, control, and planning. *International Joint Conferences on Artificial Intelligence*.
- Taylor, R. P., **Juliani, A. W.**, Bies, A. J., Boydston, C., Spehar, B., & Sereno, M. E. (2018). The implications of fractal fluency for biophilic architecture. *Journal of Biourbanism*, 6, 23-40.
- Juliani, A. W.**, Bies, A. J., Boydston, C. R., Taylor, R. P., & Sereno, M. E. (2016). Navigation performance in virtual environments varies with fractal dimension of landscape. *Journal of Environmental Psychology*, 47, 155-165.
- Juliani, A.**, Leidheiser, W., McLaughlin, A., Allaire, J., & Gandy, M. (2013). Cognitive Ability Predicts Older Adult Performance in a Complex Task but is Moderated by Social Interaction. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (Vol. 57, No. 1, pp. 1740-1744).

Preprints

- Christov-Moore, L., **Juliani, A.**, Kiefer, A., Reggente, N., Rousse, B. S., Safron, A., & Damasio, A. (2025). The Contingencies of Physical Embodiment Allow for Open-Endedness and Care. *arXiv preprint arXiv:2510.07117*.

- Juliani, A.,** Chelu, V., Graesser, L., Safron, A. (2024) A dual-receptor model of serotonergic psychedelics: therapeutic insights from simulated cortical dynamics. *Biorxiv preprint*.
- Juliani, A.,** Berges, V. P., Vckay, E., Gao, Y., Henry, H., Mattar, M., & Lange, D. (2018). Unity: A general platform for intelligent agents. *arXiv preprint arXiv:1809.02627*.

Conference Presentations

- Juliani, A.** (2025). A Machine Learning Perspective on Psychedelics and Psychopathology. *MAPS Psychedelic Science*.
- Juliani, A.,** Barnett, S., Davis, B., Sereno, M., & Momennejad, I. (2022). Neuro-Nav: A Library for Neurally-Plausible Reinforcement Learning. *The 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making*.
- Juliani, A.,** Sereno, M. (2020). A Biologically Inspired Dual Stream World Model. *NeurIPS Workshop on Biological and Artificial Reinforcement Learning*.
- Juliani, A. W.,** Yaconelli, J. P., & Sereno, M. E. (2019). Learning to Integrate Egocentric and Allocentric Information using a Goal-directed Reward Signal. *Journal of Vision*, 19(10), 162-162.
- Juliani, A.,** Bies, A., Boydston, C., Taylor, R., & Sereno, M. (2016). Spatial localization accuracy varies with the fractal dimension of the environment. *Journal of Vision*, 16(12), 1370-1370.
- Juliani, A.** (2013) Aggression Dynamics in Captive Propithecus Coquereli (2013). *Duke University Research Symposium*.

Patents

- Juliani, A. W.,** & Mattar, M. M. A. (2020). Method And System For Interactive Imitation Learning In Video Games. *U.S. Patent Application No. 16/657,868*.
- Meuleau, N. F. X., Berges, V. P. S. M., Ebrahimi, A. P., **Juliani, A. W.,** & Santarra, T. J. (2020). Method And System For A Behavior Generator Using Deep Learning And An Auto Planner. *U.S. Patent Application No. 16/660,740*.

Books & Book Chapters

- **Author**, “Simple Reinforcement Learning with Tensorflow” — Series of Medium articles (100k+ views) compiled into a book and published in Korean translation (2015 – 2017)
- **Chapter Co-Author**, “Character AI and Behaviors” — in *Created Augmented & Virtual Realities*, O’Reilly Media (2018)

Mentorship & Research Supervision

- **PhD Student Mentor**, Araya Inc. — Advised PhD interns on research design and methodology, developing structured training materials and providing individualized feedback that led to validated cognitive neuroscience models. 2020 – 2021
- **Research Internship Mentor**, Unity Technologies — Mentored summer interns in deep reinforcement learning, developing project scopes and learning milestones while providing weekly guidance on technical implementation and scientific communication. 2017 – 2019
- **Undergraduate Research Lead**, University of Oregon — Advised and mentored undergraduate research teams, developing structured protocols for behavioral experiments and coaching students in data analysis, scientific writing, and presentation skills. 2014 – 2016

Invited Tutorials & Academic Teaching

- **Reading Group Co-Organizer**, “Philosophy, Literature, & Poetry” — Online, International (2020 – Present)
- **Tutorial Lead**, “Varieties of Human-Like AI” — Cognitive Computational Neuroscience (Fall 2022)
- **Course Co-Author**, “Neural Networks” — University of Oregon (Fall 2019)
- **Tutorial Lead**, “Reinforcement Learning” — O’Reilly AI Conference, NYC & Beijing (2017, 2018)
- **Course Co-Instructor**, “Cognition” — University of Oregon (Summer 2015)
- **Teaching Assistant**, “Intro to Psychology & Cognition” — University of Oregon (2014 – 2015)

Professional Service

- **Workshop Co-Organizer**, “Learning Transferable Skills” — NeurIPS (2019)
- **Journal Reviewer**, “Deep Reinforcement Learning & Games” — IEEE Computational Intelligence Magazine (2019)

Open Source Research Software

- **Beginner’s RL Playground** (2025, GitHub, 90 stars) — An interactive web-based demonstration of fundamental tabular Reinforcement Learning (RL) algorithms in a simple grid world environment.
- **Neuro-Nav Library** (2021-2022, GitHub, 210 stars) — Offers a set of standardized environments and RL algorithms drawn from canonical behavioral and neural studies in rodents and humans.
- **Unity ML-Agents Toolkit** (2017-2020, GitHub, 18.9k stars) — Enables researchers to create and train PyTorch-based Deep RL agents within virtual environments built using the Unity Engine.
- **Obstacle Tower Environment** (2018-2019, GitHub, 549 stars) — Procedurally generated 3D environment designed to evaluate generalization in deep reinforcement learning algorithms.
- **Simple RL Tutorials** (2015-2017, GitHub, 2.3k stars) — A series of Tensorflow tutorials describing the implementation of various Deep RL algorithms including A3C and DQN.